Continued…
The Threat Environment: Attackers and Their Attacks

Reference:

Corporate & Network Security, Chapter 1

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Agenda

- The threat environment—attackers and their attacks
- Basic security terminology
- Employee and ex-employee threats
- Traditional external attackers
- The criminal era and competitor threats
- Cyberwar and cyberterror

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2.3 Traditional External Attackers

- Traditional external attackers use the Internet to send malware into corporations, hack into corporate computer and do other damage

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2.3.1 Classic Malware: Viruses and Worms

- **Malware**
  - A generic name for any "evil software"

The most widely known type of malware is the virus, but there are many other types including worms, trojan horses, RATs, spam, and others

The first external malware attackers were malware writers

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Classic Malware: Viruses and Worms

- **Malware**

Malware is a very serious threat. In June 2006 Microsoft reported results from a survey of users who allowed their computers to be scanned for malware. The survey found 16 million pieces of malware on the 5.7 million machines examined.

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Classic Malware: Viruses and Worms

- **Viruses**
  - Programs that attach themselves to legitimate programs on the victim's machine
  - Later when infected programs are transferred to other computers and run, the virus attaches itself to other programs on those machines

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Classic Malware: Viruses and Worms

- **Viruses**
  - Spread today primarily by e-mail with infected attachments
  - Also by instant messaging, file transfers, file sharing programs, downloads from malicious websites, etc

Through networked applications, viruses can spread very rapidly today

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Classic Malware: Viruses and Worms

- **Example**

When Macintosh users searched BitTorrent sites in early 2009, they found that they were able to download the newly released CS4. They would also download a program on the downloaders computer to crack the software. The copy of CS4 was clean, however, when the downloader ran the cracking program, he or she got a dialog box saying "Adobe CS4 Crack [intel] requires that you type your password."

-"Mac Trojan Horse Found In Pirated Adobe CS4 (http://blogs.zdnet.com)

Classic Malware: Viruses and Worms

- **Worms**
  - Full programs that do not attach themselves to other programs
  - Like viruses, can spread by e-mail, instant messaging, and file transfers

In general worms act much like viruses and can spread via email and in other ways that viruses spread

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Classic Malware: Viruses and Worms

- **Direct-propagation Worms**
  - In addition, direct-propagation worms can jump from one computer to another without human intervention on the receiving computer
  - Computer must have a vulnerability for direct propagation to work
  - Direct-propagation worms can spread extremely rapidly because they do not have to wait for users to act

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Classic Malware: Viruses and Worms

- **Direct-propagation Worms**

Direct-propagation worms jump directly to computers that have vulnerabilities; they then use these computers to jump to other computers (all without any action on the user's part)

Researchers at UC Berkeley: a worst case direct-propagation worm could do USD 50 billion damage to the US alone (http://www.techweb.com)

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Classic Malware: Viruses and Worms

- **Slammer Worm**

On January 25, 2003, the Slammer worm exploded across the Internet. In ten minutes, before a handful of people knew it existed, Slammer had infected 90% of all vulnerable computers on the entire Internet. Although Slammer did not erase hard-disks or do other damage, it caused massive damage by spreading so quickly it choked parts of the Internet. Around the world, ATMs became unusable, police departments lost their ability to communicate and most users in Korea lost their service.

-Raymond R. Panko, 2003

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Classic Malware: Viruses and Worms

- **Blended Threats**
  - Malware propagates in several ways—like worms, viruses, compromised webpages containing mobile code, etc.

By propagating in multiple ways, blended threats increase their likelihood of success.

MessageLabs reported in August 2006 that 1% of all email contains viruses, worms, or blended threats. During major outbreaks, one in ten email messages may contain viruses, worms, or blended threats.
(http://www.messagelabs.com)

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Classic Malware: Viruses and Worms

- **Payloads**
  - Pieces of code that do damage
  - Implemented by viruses and worms after propagation
  - Malicious payloads are designed to do heavy damage

Benign payloads merely pop up a message on the user's screen or do some other annoying but nonlethal damage

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Classic Malware: Viruses and Worms

- **Payloads**

Malicious payloads can do extreme damage, for example, by randomly deleting files from the victim's hard disk drive or by installing some other types of malware

Virus and worm payloads also frequently soften up the computer by disabling its antivirus software and by taking other actions that leave it highly vulnerable to subsequent virus and worm attacks

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Classic Malware: Viruses and Worms

In 2004, the Aberdeen group surveyed 162 companies. They found that each firm lost an average of USD 2 million per virus or worm incident and spent an additional USD 100,000 to clean up computers after an attack. Both numbers increased with company size. Most companies reported enduring on average one incident per year, although many firms reported multiple incidents. (http://www.aberdeen.com)

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Classic Malware: Viruses and Worms

Another security firm Mi2g estimated that damage from malware in 2004 alone averaged USD 290 per PC in the firms it studied.

(http://www.mi2g.com)

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2.3.2 Trojan Horses and Rootkits

- **Nonmobile Malware**
  - Must be placed on the user's computer through one of a growing number of attack techniques

1. Placed on computer by hackers
2. Placed on computer by virus or worm as part of its payload
3. The victim can be enticed to download the program from a website or FTP site
4. Mobile code executed on a webpage can download the nonmobile malware

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Trojan Horses and Rootkits

- **Trojan Horses**
  - A program that replaces an existing system file, taking its name

Most nonmobile malware programs are trojan horses

Early trojan horses were programs that pretended to be one thing, such as a game or a pirated version of a commercial program, but really were malware. Many of these classic trojan horses still exist.

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Trojan Horses and Rootkits

- **Trojan Horses**

Today, however, when we talk about a trojan horse, we mean a program that hides itself by deleting a system file and taking on the system file's name.

Trojan horses are difficult to detect because they look like legitimate system files

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Trojan Horses and Rootkits

- **Trojan Horses**
  - Remote Access Trojans (RATs)
    - Remotely control the victim's PC

The attacker can remotely do pranks such as opening and closing your CD drive, or by typing text on your screen

There are many legitimate remote access programs that allow a remote user to work on a machine or do diagnostics. However, RATs are typically stealthy in order to avoid detection by the owner of the machine

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Trojan Horses and Rootkits

- **Trojan Horses**
  - Downloaders
    - Small Trojan horses that download larger Trojan horses after the downloader is installed

Downloaders are usually fairly small programs which makes detection difficult. However the larger trojan horse is capable of doing much more damage

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Trojan Horses and Rootkits

- **Trojan Horses**
  - **Spyware**
    - Programs that gather information about you and make it available to the adversary

Although the biggest problem with spyware is the theft of information, spyware also tends to make computers run sluggishly

One new form of spyware is *camera spyware* which spies on the victim visually by turning on its camera and perhaps also its microphone

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Trojan Horses and Rootkits

- **Trojan Horses**
  - **Spyware**
    - Cookies that store too much sensitive personal information

Websites are allowed to store small text strings called *cookies* on your PC. The next time you go to the website the website can retrieve the cookie. Cookies have many benefits like remembering your password each time you visit. Cookies can also remember what happened last in a series of screens leading to purchases. However, when cookies record too much sensitive information about you, they become spyware.

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Trojan Horses and Rootkits

- **Trojan Horses**
  - **Spyware**
    - Keystroke loggers

Keystroke loggers capture all of your keystrokes. They then look through the collected keystrokes for usernames, passwords, credit card numbers, and other sensitive information. They send this information to the adversary.

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Trojan Horses and Rootkits

- **Trojan Horses**
  - **Spyware**
    - Password-stealing spyware

Tells you that you have been logged out of the server you are visiting and asks you to retype your username and password. If you do the spyware sends the username and password to the attacker.

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Trojan Horses and Rootkits

- **Trojan Horses**
  - **Spyware**
    - Data mining spyware

Searches through your disk drives for the same types of information sought by keystroke loggers. It also sends this information to the adversary.

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